

Homework 2**Due: Saturday, September 12 on Canvas**

1. (10 points) The PMF of a binomial random variable X is given by

$$p_X[k] = \binom{M}{k} p^k (1-p)^{M-k} \text{ for } k = 0, 1, \dots, M$$

Consider the limit $M \rightarrow \infty$ and $p \rightarrow 0$ but $pM = \lambda = \text{constant}$, and express $p_X[k]$ in terms of the parameter λ . What PMF is this?

2. (10 points) Let X be a Poisson random variable with parameter λ . Define the transformation $Y = g(X)$ as

$$Y = \begin{cases} 1 & \text{if } X = k \text{ is even} \\ -1 & \text{if } X = k \text{ is odd} \end{cases}$$

Find the PMF of Y .

3. (10 points) Say one wishes to predict the outcome of a coin-tossing experiment through the best estimate of the value of a Bernoulli random variable with parameter $0 \leq p \leq 1$. Let this estimated value be b .

- (4 points) Find b .
- (2 points) What is the mean-square error of this estimate? Express the error in terms of the parameter p .
- (4 points) For what value of p does the error reach maximum? Why?

4. (15 points) Consider the binomial random variable X_b .

- (3 points) Calculate its characteristic function.
- (6 points) Determine the expected value and variance from the characteristic function.
- (6 points) Table 6.1 lists the characteristic functions of a number of discrete random variables. Show that the characteristic function of the binomial random variable X_b becomes that of the Poisson random variable X_p in the limit $M \rightarrow \infty$ and $p \rightarrow 0$ ($pM = \lambda = \text{constant}$).

5. (5 points) For a PSK digital communication system, the probability of error is given by

$$P_e = P \left[\frac{A}{2} + W \leq 0 \right],$$

where $W \sim \mathcal{N}(0,1)$. Prove $P_e = Q\left(\frac{A}{2}\right)$.

6. (5 points) Consider $X \sim \mathcal{N}(0,1)$ and the affine transformation $Y = \sqrt{\sigma^2}X + \mu$. Prove $Y \sim \mathcal{N}(\mu, \sigma^2)$.

7. (5 points) Let $X \sim \mathcal{N}(0,1)$ and $Y = X^2$. Find $p_Y(y)$.

8. (10 points) Let X be a Rayleigh random variable. Consider the transformation:

$$Y = g(X) = \begin{cases} X, & 0 \leq X < x_{\max} \\ x_{\max}, & X > x_{\max} \end{cases}$$

Find $p_Y(y)$.

9. (10 points) Let X be an exponential random variable.

- (2 points) Calculate its characteristic function $\phi_X(\omega)$
- (8 points) Calculate the n th moment $E[X^n]$ using the characteristic function.

10. (20 points) Consider a Gaussian random variable: $X \sim \mathcal{N}(3,1)$.

- (5 points) Express the probability $P[X > 8]$ as an integral. As an order of magnitude estimate, how many realizations of this random variable are needed to have 10 events of $X > 8$?

- (b) (5 points) For an exponential random variable Y with the PDF: $p_Y(y) = e^{-(y-3)}U(y-3)$, where $U(y)$ is the unit step function, how many realizations are needed to have 10 events of $Y > 8$?
- (c) (10 points) Using the principle of importance sampling, express the probability $P[X > 8]$ as the expected value of some function of the random variable Y in terms of its own PDF. Find an explicit expression of this function.